Project Documentation & Business Proposal

Project Name: FCAI E-Club

Prepared for: Eng. Mohamed Mohamed Hassan Fahmy

Prepared By: Eng. Ahmed Wael Mohamed

Date: August 16, 2026

Version: v1.0

PART 1: EXECUTIVE BUSINESS PROPOSAL

EXECUTIVE SUMMARY

FCAI E-Club is a non-profit student-led organization bridging the gap between academic theory and industry reality for engineers and entrepreneurs. Currently operating without a centralized digital footprint, the club requires an online platform to establish its brand identity, manage its growing membership, showcase its operational impact, and facilitate its extensive calendar of technical workshops and events.

This proposal outlines the development of a comprehensive web application that serves as the central nervous system for the E-Club. By investing in this platform, FCAI E-Club will professionalize its operations, attract high-value industry partners, increase student participation, and create a scalable digital infrastructure for future growth.

PROBLEM STATEMENT

- Lack of centralization: Currently, there's no source of truth for club information, event schedules, or member achievements.

- Operational Inefficiency: Manual tracking of members, workshops, and partnerships is time-consuming and subjected to error.
- Limited Visibility: The club lacks a professional public-facing presence to showcase its achievements and attract top-level industry partners and ambitious students

PROPOSED SOLUTION

We propose the development of a fully responsive, modern web application built on a high-performance tech stack. The platform will feature a public-facing informational portal and a secure, authenticated back-end for members and administrators.

- Brand Authority: Establishing a good strong digital identity that shows both vision and mission effectively.
- Operational Hub: A centralized system for logging club history, tracking KPIs across different technical and non-technical committees.
- Engagement Engine: A dynamic portal for listing events, workshops, and bootcamps, and track registrations throughout the club's activity.
- Partnership Gateway: A dedicated section to attract and formalize relationships with industry leaders.

EXPECTED ROI & BENEFITS

- Increased Membership: A professional, easy-to-use registration process will drive student sign-ups.
- Corporate Growth: A polished digital showcase will facilitate easier acquisition of industry partners (e.g., Flat6Labs, Divenore).
- Time Savings: Automation of workshop registrations and member management will drastically reduce administrative overhead for club leaders.
- Institutional Knowledge Retention: The platform will serve as a permanent, searchable archive of club projects, workshops, and past activity log.

PART 2: FUNCTIONAL REQUIREMENTS & TECH SPECIFICATIONS

To ensure scalability, performance, and a modern user experience, we recommend the following stack:

- **Frontend (UI/UX):** Next.js (React) framework for SEO performance and dynamic routing. Styled with Tailwind CSS for rapid, responsive design.
- **Backend API & Database:** Node.js with Express, connected to a PostgreSQL and Supabase database to manage relational data (users, events, registrations, contributions).
- **Authentication:** NextAuth for secure member login.

SITE ARCHITECTURE & KEY PAGES

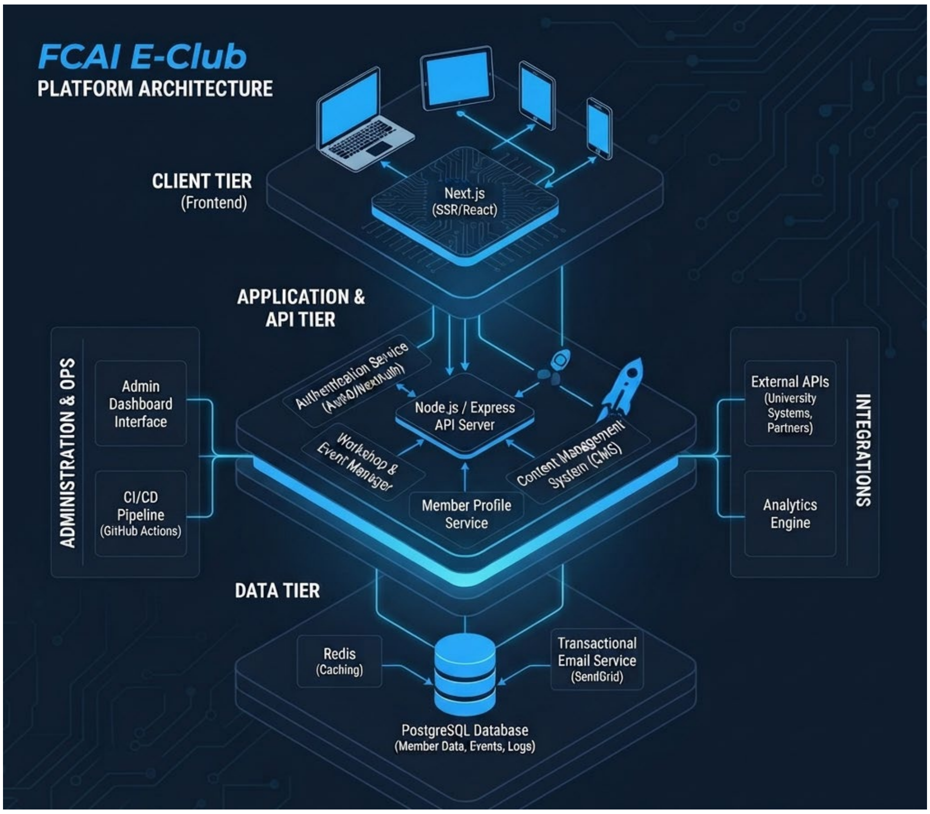

Figure 1: FCAI E-Club Platform Architecture

GLOBAL ELEMENTS

- **Navigation Bar (Header):**
  - Logo (Left-aligned, "FCAI E-Club Logo Image")
  - Navigation Links: About, Partnerships, Team, Workshops
  - CTA Button: "Register" (Directs to sign-up flow)
- **Footer:**
  - Logo
  - Quick Links (Privacy Policy, Terms of Service, Contact us, FAQ)
  - Social Links (Facebook, Instagram, LinkedIn)
  - Copyright in the bottom

PUBLIC-FACING PAGES (NON-AUTHENTICATED USERS)

- **Landing Page (Hub):**
  - **Hero Section:** Dynamic headline ("Engineering the Future" or "Where tech meets innovation"), introductory paragraph, and two primary CTAs ("Join the Club", "Explore Events").
  - **Impact Statistics:** Three data cards (Active Members, Workshops Hosted, Industry Partners, Events Held).
  - **About Us Snippet:** Brief intro to the club's mission of incubating technical founders, linking to the full 'About' page.
  - **Upcoming Events:** A scrollable list of the next two upcoming workshops (showing date, title, description, location, time) with a "View All" link.
  - **Backed by Industry Leaders:** A logo ticker/grid of partner companies.

- **About Page (Organizational Deep Dive):**
  - **Hero Section:** "Engineering the Future" with mission statement.
  - **Mission Section:** Text and a high-res visualization (Image 1).
  - **Vision Section:** Text and an abstract tech network visualization (Image 2).
  - **Operation Log (Club Timeline):** An interactive vertical timeline detailing club history. Each point contains a title, date, and description.

- **Partnerships Page:**
  - **Hero:** "Our Ecosystem Partners".
  - **Partner Logos Grid:** Display of existing partner logos (placeholder boxes in design).
  - **"Join Our Network" Section:** Pitch text outlining benefits (Access to student talent, Brand visibility, Collaborative projects).
  - **Contact Form:** Input fields for Company Name, Contact Email, Partnership Interest (Dropdown: Sponsorship, Mentorship, Recruitment), and a "Submit Proposal" button.
- **Team Page (Meet the Visionaries):**
  - **Hero:** "Meet the Visionaries". Intro paragraph.
  - **Committees Section:** Cards for Technical, Marketing, and Logistics committees, showing member count and a "View Team" CTA.
  - **All Members Section:** A filterable/tabbed grid (All, Technical, Marketing) of all members. Each member card must display: Profile Photo, Name, Title/Role (e.g., Tech Lead), and Primary Committee affiliation.
- **Workshops & Events Listing (Main Catalog):**
  - **Hero:** "Workshops & Bootcamps".
  - **Filters:** Tabs for All Tracks, Technical, Soft Skills, Business.
  - **Event Cards:** Cards for individual events (e.g., Advanced React Patterns). Each card must display: Status (Open, Closed), Date (MMM DD, YYYY), Title, Description, Track Tag, and a "Read More" CTA.
- **Individual Event Detail (e.g., Advanced Prompt Engineering):**
  - **Hero:** Event Title, brief description, and "Upcoming Workshop" tag.
  - **About the Workshop:** Detailed text description.
  - **Instructor Profile:** Instructor Photo, Name, Title.

- **Curriculum Timeline:** Vertical list showing specific times and topics (e.g., 10:00 AM - Foundations).
- **Workshop Details Sidebar:** Card displaying Date, Time, Location (with "Virtual Link Provided on Registration" note), and Capacity remaining (e.g., 50 Seats, 24 Remaining).
- **Registration Form (Secure):** Email input field and a "Register Now" button.

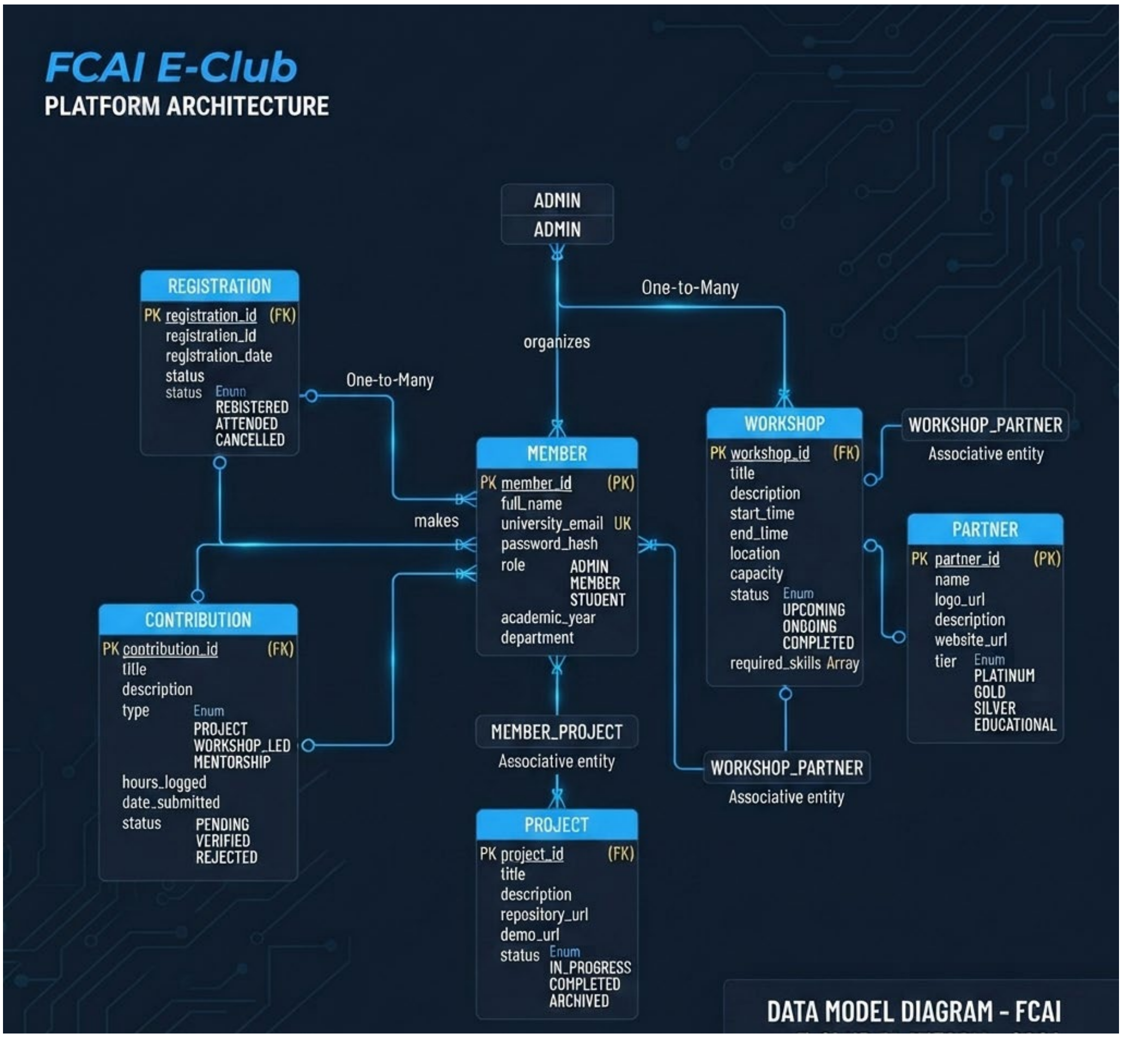

Figure 2: E-Club Platform Architecture (Data Model Diagram)

PART 3: DEVELOPMENT AND DEPLOYMENT ROADMAP

Phase 1: Discovery & Planning

- Finalize database schema (Prisma) based on the entities identified above.
- Set up GitHub repository, CI/CD pipeline, and development environments.
- Establish project management board (GitHub Projects/Notion) for task tracking.

Phase 2: Core Infrastructure & Public Site

- Setup Next.js project, integrate Tailwind CSS, and configure routing.
- Develop global Nav and Footer components.
- **Develop Public Pages:** Landing, About (including static timeline and KPI matrix visuals), Partnerships (including form handling), and Team (including filtering logic).
- Implement basic SEO meta-tags for public pages.

Phase 3: Events & Registration Engine

- Develop the Workshops Catalog and individual Event Detail pages.
- Implement the Registration Form logic (saving to DB).
- **Email Automation:** Integrate transactional email service to trigger confirmation emails upon registration.
- Develop the Member Profile template.

Phase 4: Authentication & Member Dashboard

- Implement user Authentication (Login/Register flow).

Phase 5: Admin Portal & Content Management

- **Data Management Interface:** Build the admin interface to allow authorized users to:
  - CRUD (Create, Read, Update, Delete) Events and Workshops.
  - Update the "Workshop Matrix" KPI percentages in the About section.
  - Manage user roles and committee assignments.
  - Export workshop registration lists (CSV).

Phase 6: Testing & Deployment (Week 15)

- **QA Testing:** Cross-browser compatibility checks, mobile responsiveness verification on all pages, and form validation testing.
- **Security Audit:** Ensure authentication and API routes are secure.
- **UAT:** User Acceptance Testing with key club stakeholders.
- **Production Deployment:** Deploy frontend to Vercel (recommended for Next.js) and use Supabase as a stable backend/database provider.

Phase 7: Handover & Documentation

- Provide administrative training session for club high-board members on using the CMS.
- Deliver technical documentation of the codebase and API.

CLOSING REMARKS

The FCAI E-Club platform represents a pivotal milestone in modernizing how student-led tech organizations operate, scale, and engage with their communities. By bridging robust full-stack architecture with a streamlined user experience, this documentation and blueprint establish a solid foundation for sustainable growth, efficient event management, and impactful industry collaboration.

- **Scalable Technical Foundation:** Built upon a high-performance Next.js and Node.js stack backed by a relational PostgreSQL database, ensuring long-term maintainability and speed.
- **Centralized Operational Hub:** Unifies member profiles, contribution logs, workshop tracking, and administrative controls into a single, cohesive digital ecosystem.
- **Enhanced Ecosystem Growth:** Strengthens visibility for corporate partners while empowering students with a professional platform to showcase their technical projects and milestones.

As the project transitions from architectural planning and database modeling into active development sprints, this platform will serve as the digital backbone for the FCAI Entrepreneurship Club empowering the next generation of student engineers and founders to build, innovate, and lead with confidence.